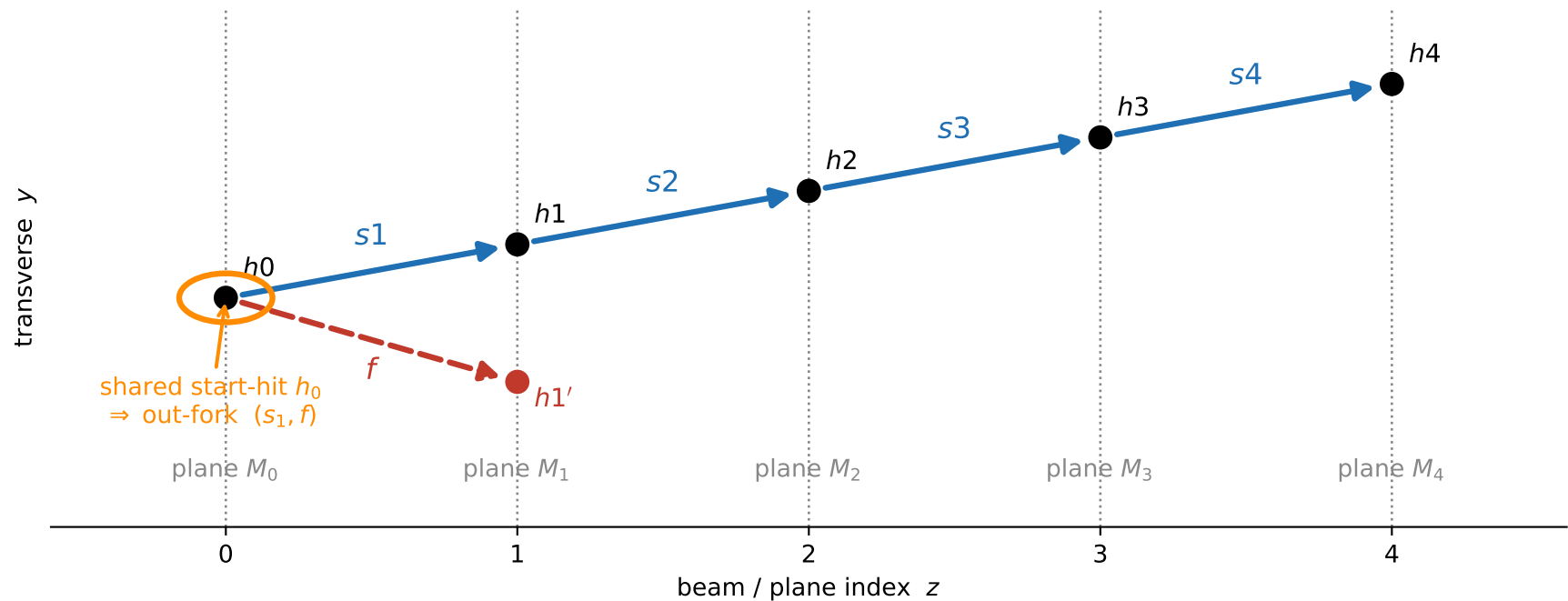
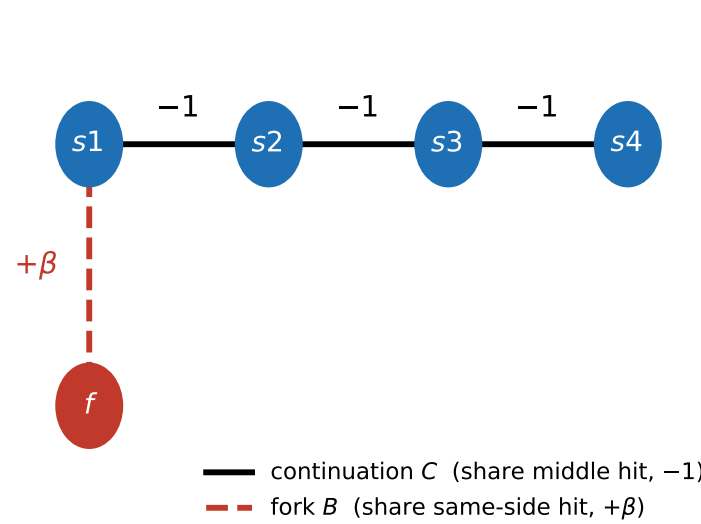


Bifurcation term — from labeled picture to Hamiltonian matrix

(a) Detector schematic: a true 4-segment chain $s_1-s_2-s_3-s_4$ and a false fork f sharing s_1 's start-hit



(b) The two graphs the picture defines



(c) $A' = (\gamma + \delta)I - C + \beta B$ (here $\gamma + \delta = 4$)

	s_1	s_2	s_3	s_4	f
s_1	$\gamma + \delta$	-1	0	0	$+\beta$
s_2	-1	$\gamma + \delta$	-1	0	0
s_3	0	-1	$\gamma + \delta$	-1	0
s_4	0	0	-1	$\gamma + \delta$	0
f	$+\beta$	0	0	0	$\gamma + \delta$

$\mathbf{b}' = \delta \mathbf{1}$ (off-diagonal form); full form: $\text{diag} \rightarrow \gamma + \delta + 2\beta$, $\mathbf{b}'' = (\delta + \beta) \mathbf{1}$